JATIN KUMAR SINHA

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Professional Summary

Computer Science and Engineering student at Lovely Professional University seeking a Software Developer Internship. Built full-stack and backend projects with Java, Spring Boot, FastAPI and React, including a Redis-like TCP server with RESP2 parsing, persistence and primary-replica replication.

Technical Skills

Languages: Java, Python, JavaScript, SQL
Backend and APIs: Spring Boot, FastAPI, REST APIs, API Integration, JWT Authentication, WebSockets, WebClient
Frontend: React, Vite, HTML, CSS, Tailwind CSS, Axios
Databases and Caching: PostgreSQL, Redis, Caffeine Cache
Networking and Protocols: TCP/IP, Java Sockets, RESP2, Java I/O, Virtual Threads
Tools: Docker, Git, GitHub, Vercel, Render, Railway
Core Concepts: Object-Oriented Programming, Multithreading, Client-Server Architecture, API Design, Caching

Projects

Mini Redis in Java | October 2026

Tech: Java 21, TCP Sockets, RESP2, Java I/O, Snapshots, Virtual Threads, Git/GitHub
GitHub: github.com/Jatinkumarsinha4112/Mini-Redis-Java

- Built a Redis-like TCP server in Java 21 that parses RESP2 and supports string, list, counter and key-expiration commands.
- Implemented persistent snapshots with restart recovery, plus primary-replica synchronization with write propagation and acknowledgement offsets.
- Verified command handling, persistence, read-only replica behavior and replication recovery with assertion-based integration checks.

F1 Intelligence Hub | June 2026

Tech: Python, FastAPI, React, JavaScript, Vite, Axios, Jolpica API, Render, Vercel
Live demo: f1-intelligence-hub-tan.vercel.app

- Built a full-stack Formula 1 dashboard with a React/Vite frontend and FastAPI backend, using the Jolpica API for season-specific driver standings, constructor standings and race schedules.
- Added season selection across the dashboard, driver profiles and driver comparisons, plus a historical champions archive.
- Implemented asynchronous API requests, expiring response caching and loading/error states; deployed on Vercel (frontend) and Render (backend).

BallKnowledge | May 2026

Tech: Java, Spring Boot, REST APIs, Caffeine Cache, Docker
Github: github.com/Jatinkumarsinha4112/-ballknowledge

- Built a football analytics platform for live match tracking, league standings and player statistics using real-time APIs.
- Built a layered Spring Boot backend with DTO handling, centralized exception handling, Caffeine caching and Redis-compatible configuration.

Real-Time AQI Alert System | March 2026

Tech: Java, OpenWeather API, JSON Parsing, JavaMail API, SMTP
Github: github.com/Jatinkumarsinha4112/AQI-Alert-System-using-Java

- Built a Java app that fetches real-time AQI data from the OpenWeather API and flags unsafe pollution conditions.
- Implemented JSON parsing, AQI analysis logic and automated email alerts using the JavaMail API and SMTP.

Training

AI and Machine Learning Summer Training, Lovely Professional University | June 2025 – July 2025
Tech: Python, NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, NLP

- Completed applied AI/ML training; built supervised and unsupervised models with Linear Regression, KNN, Decision Trees and Clustering, and practiced preprocessing and visualization.
- Developed an interactive mental-health chatbot with text preprocessing, tokenization and sentiment analysis.

Certifications

- Oracle Cloud Infrastructure Certified AI Foundations Associate | August 2026 | Certificate: View certificate
- Oracle AI Database Certified Foundations Associate | August 2026 | Certificate: View certificate
- ChatGPT-4 Prompt Engineering: ChatGPT, Generative AI and LLM | August 2025 | Certificate: View certificate
- Computational Theory: Language Principle and Finite Automata Theory | August 2025 | Certificate: View certificate

Education

Bachelor of Technology, Computer Science and Engineering | August 2023 – Present
Lovely Professional University, Phagwara, Punjab | CGPA: 6.73

Intermediate (Class XII), PCM | March 2021 – May 2022
Bishop Scott Boys' School, Patna, Bihar | 72%

Matriculation (Class X) | March 2019 – May 2020
Bishop Scott Boys' School, Patna, Bihar | 75%